

Once we've collected our data, we can plot a histogram of the change in blood pressure after four weeks on the drug for the treatment and control groups. We might see something like the results in figure 7.2.

Figure 7.2. Histogram of the measured blood pressure change for the control and treatment groups in a simulated randomized control trial. The x-axis is the change in blood pressure from the start (before treatment) to after treatment. We want blood pressure to decrease, so negative numbers are good. We count the number of patients who have each value of blood pressure change, so the peak at –3 for the control group means that most of those patients had a blood pressure drop of 3 mmHg. You can see that there are two subgroups of patients in the treatment group: one group had a significant reduction in blood pressure, and another group had minimal to no effect. We call this a bimodal distribution, where mode is another word for "peak" in the distribution.